

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA17	Course Title:	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	35% of CIA	Duration / Marks:	60 mins
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
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Section / Batch:	B.Tech-IT	Date:	05/10/2026

Code of Conduct

I Dinesh R certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: R Dinesh

Instructions

- Read each question carefully before answering.
- Answer all four questions. Show every step: state your assumptions, the method you choose and why, your calculations, and finally interpret the result for the agent or AI system in the problem.
- Questions are drawn from Foundations and History of AI, Intelligent Agents (Agents and Environments) and the Concept of Rationality.
- Prepare digital Write up units and clear final answers (marks, minutes, percentage points, expected utility); present tables and working neatly.
- Marks will be awarded using the Analytical Problem Solving rubric: problem understanding, method selection, solution process, accuracy of calculation and interpretation of results.
- Upload in Github and GCR (Question paper with regNo, Name, Answers and rubrics and evaluation page at last).
- Duration: 1 hour (15 minutes per question).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Definitions of AI (four approaches), Turing Test, eras and milestones in the history of AI	Q3
2. Intelligent Agents: Agents and Environments	Agent function and program, PEAS description, task environment properties, agent architectures	Q1, Q2
3. Concept of Rationality	Performance measure, percept sequence, expected utility, rational versus omniscient agent	Q1, Q4

Question 1: Two-Room Vacuum World

Topic: Agents and environments, concept of rationality | 15 minutes

A vacuum-cleaner agent lives in a world with two squares, A and B. At each time step it perceives its own location and whether that square is Dirty or Clean. Its actions are Suck, Right, Left and NoOp. The agent starts in square A, and both squares are initially Dirty. Once a square is cleaned, dirt never reappears.

Performance measure over a lifetime of $T = 6$ time steps: +1 for every clean square at the end of each time step, and a penalty of p for every Left or Right move (take $p = 1$ unless told otherwise).

- **Agent X (simple reflex):** if the current square is Dirty, Suck; otherwise move to the other square.
- **Agent Y (reflex with internal state):** if the current square is Dirty, Suck; otherwise, if it remembers that both squares have been cleaned, NoOp; otherwise move to the other square.

- (a) Write the PEAS description (Performance measure, Environment, Actuators, Sensors) of this agent from the data given. **[4 Marks]**
- (b) Trace Agent X for all 6 time steps in a table (action, clean squares at end of step, move penalty, step score) and find its total score for $p = 1$. **[5 Marks]**
- (c) Trace Agent Y in the same way and find its total score for $p = 1$. **[4 Marks]**
- (d) Derive the total score of X and of Y as functions of p . For which values of p is Y strictly better than X? **[3 Marks]**
- (e) Which agent is rational? Interpret your result: explain how the performance measure and the percept history decide rationality, and why X cannot match Y even though both see the same percepts at each step. **[4 Marks]**

(a) PEAS Description

PEAS element	Description
Performance measure	+1 for every clean square at the end of each time step; $-p$ for every Left/Right move.
Environment	Two squares A and B; initially both Dirty; dirt does not reappear.
Actuators	Suck, Right, Left, NoOp.
Sensors	Perceives current location and whether the current square is Dirty or Clean.

(b) Agent X – Simple Reflex

Step	Location / percept	Action	Clean squares at end	Move penalty	Step score
1	A Dirty	Suck	A, B = Dirty/Clean → 1 clean	0	1
2	B Clean	Right → A	A clean	$-p$	$1-p$
3	B Dirty	Suck	A and B clean	0	2

4	A Clean	Left $\rightarrow$ B	A and B clean	$\neg p$	$2-p$
5	B Clean	Right $\rightarrow$ A	A and B clean	$\neg p$	$2-p$
6	A Clean	Left $\rightarrow$ B	A and B clean	$\neg p$	$2-p$

For $p = 1$: total score = $1 + 0 + 2 + 1 + 1 + 1 = 6$.

Therefore, Agent X total score = 6 for $p = 1$.

(c) Agent Y – Reflex with Internal State

Step	Location / percept	Action	Clean squares at end	Move penalty	Step score
1	A Dirty	Suck	A clean	0	1
2	B Clean	Right $\rightarrow$ B	A clean	$\neg p$	$1-p$
3	B Dirty	Suck	A and B clean	0	2
4	A/B clean remembered	NoOp	A and B clean	0	2
5	A/B clean remembered	NoOp	A and B clean	0	2
6	A/B clean remembered	NoOp	A and B clean	0	2

For $p = 1$: total score = $1 + 0 + 2 + 2 + 2 + 2 = 9$.

Therefore, Agent Y total score = 9 for $p = 1$.

(d) Total scores as functions of p

Agent X makes four moves (steps 2, 4, 5 and 6), so:

$$S_x(p) = 6 - 4p$$

Agent Y makes only one move (step 2), so:

$$S_y(p) = 9 - p$$

Y is strictly better when:

$$9 - p > 6 - 4p \rightarrow 3 + 3p > 0 \rightarrow p > -1.$$

Since a move penalty is normally $p \geq 0$, Agent Y is strictly better for every physically meaningful non-negative penalty p .

(e) Rationality and interpretation

Agent X cannot match Y because X is a simple reflex agent: it uses only the current percept. Once a clean square is perceived, it moves, even when both squares are already clean. Agent Y uses percept history/internal state to avoid unnecessary moves and their penalties.

Question 2: Task Environment Analysis and Agent Selection

Topic: Agents and environments (PEAS, environment properties, agent types) | 15 minutes

Three AI agents are to be deployed in the following task environments:

- **E1:** An autonomous taxi driving through city traffic.
- **E2:** A part-picking robot that uses a camera to pick defective parts off a moving conveyor belt and place them in bins.
- **E3:** A chess-playing agent in a tournament game played with a clock.

Difficulty score convention: for each of the six properties below, the harder value scores 1 point and the easier value scores 0. A semi-dynamic environment scores 0.5 on the Static/Dynamic property.

Property	Easier value (0)	Harder value (1)
Observability	Fully observable	Partially observable
Number of agents	Single agent	Multi-agent
Determinism	Deterministic	Stochastic
Episodic / Sequential	Episodic	Sequential
Static / Dynamic	Static	Dynamic (semi-dynamic = 0.5)
Discrete / Continuous	Discrete	Continuous

- (a) Classify each environment on all six properties with a one-phrase justification for each. **[9 Marks]**
- (b) Compute the difficulty score of E1, E2 and E3 and rank them from hardest to easiest. **[4 Marks]**
- (c) For the hardest and the easiest environment, choose the most suitable agent architecture (simple reflex, model-based reflex, goal-based, utility-based, learning) and justify the choice using your classification. **[4 Marks]**
- (d) Interpret: explain how the properties of an environment, not just its domain, decide how complex the agent program must be. **[3 Marks]**

(a) Classification of E1, E2 and E3

Environment	Observability	Agents	Determinism	Episodic / Sequential	Static / Dynamic	Discrete / Continuous
E1 – Autonomous taxi	Partially observable – driver/agent cannot observe every	Multi-agent – other vehicles/pedestrians interact	Stochastic – traffic and disruptions are uncertain	Sequential – current decisions affect later states	Dynamic – traffic changes while acting	Continuous – position, speed and time vary continuously

	traffic condition					
E2 – Part-picking robot	Partially observable – camera has limited visual information	Single agent – one robot performs the picking task	Stochastic – detection/picking outcomes can vary	Episodic – each pick is largely an individual task	Dynamic – conveyor and parts keep moving	Continuous – motion and positions vary continuously
E3 – Chess with clock	Fully observable – complete board state is visible	Multi-agent – two competing players	Deterministic – legal moves determine board transitions	Sequential – every move changes the future game	Semi-dynamic – board is static while thinking, but clock changes	Discrete – finite board squares and moves

(b) Difficulty score and ranking

Property	E1 Taxi	E2 Robot	E3 Chess
Partially observable	1	1	0
Multi-agent	1	0	1
Stochastic	1	1	0
Sequential	1	0	1
Dynamic / semi-dynamic	1	1	0.5
Continuous	1	1	0
Total difficulty score	6	4	2.5

Ranking from hardest to easiest: E1 Taxi (6) > E2 Part-picking robot (4) > E3 Chess (2.5).

(c) Suitable agent architectures

Environment	Recommended architecture	Justification
Hardest – E1 Taxi	Learning agent	The taxi faces partial observability, multiple agents, stochastic

		outcomes, sequential decisions, dynamic traffic and continuous control. Learning can improve decisions from experience and adapt to changing traffic conditions.
Easiest – E3 Chess	Utility-based agent	Chess is fully observable and discrete, but it is multi-agent and sequential. A utility function can compare winning/losing positions and also account for the tournament clock, allowing the agent to select the action with the best expected outcome.

(d) Interpretation

The domain name alone does not determine agent complexity. Environment properties determine what information the agent receives, how predictable the result is, whether other agents act, whether decisions affect future states, whether the world changes during computation, and whether control is continuous. Therefore, the more difficult properties an environment has, the more sophisticated the agent program must be. E1 needs substantially more sophisticated decision-making than E3 because it combines all six challenging properties.

Question 3: Turing Test Data Across the History of AI

Topic: Foundations and history of AI (Turing Test, acting humanly, eras of AI) | 15 minutes

In a Turing-test style study, judges held five-minute text conversations with an unknown party and then said whether it was a human or a machine. Each system below faced 50 independent interrogations in which it was the machine. The table shows how many times the judge correctly identified it as a machine.

System	Era / technique	Interrogations	Correctly identified
ELIZA	1960s, keyword pattern matching	50	47
Chatbot B	2010s, scripted rules plus statistical NLP	50	36
System C	2020s, large neural language model	50	29

Reference: Turing (1950) predicted that an average interrogator would have no more than a 70% chance of making the right identification after five minutes. A judge who is purely guessing is right 50% of the time.

- (a) Compute the correct-identification rate and the fooling rate (percentage of interrogations in which the machine was taken for a human) for each system. **[6 Marks]**
- (b) Decide which systems satisfy Turing's 70% criterion. State the method you use and how far each system is from the 50% chance level. **[5 Marks]**
- (c) Compute the improvement in fooling rate from ELIZA to System C in percentage points and as a ratio. Also compute the improvement from Chatbot B to System C in percentage points. **[5 Marks]**
- (d) Interpret: the Turing Test measures "acting humanly". Explain why a high fooling rate does not prove that System C is thinking or acting rationally, using one idea from the foundations of AI (for example the four approaches to defining AI, or the Chinese Room argument). **[4 Marks]**

(a) Correct-identification rate and fooling rate

Formula: Correct-identification rate = (correct identifications / 50) × 100. Fooling rate = 100 – correct-identification rate.

System	Correctly identified	Correct-identification rate	Fooling rate
ELIZA	47/50	94%	6%
Chatbot B	36/50	72%	28%
System C	29/50	58%	42%

Example for System C: Correct rate = $(29/50) \times 100 = 58\%$; fooling rate = $(50 - 29)/50 \times 100 = 42\%$.

(b) Turing 70% criterion and distance from 50% chance

Method: compare each correct-identification rate with the 70% reference criterion and with the 50% guessing level.

System	Correct rate	Meets 70% criterion?	Distance from 50% chance
ELIZA	94%	Yes	$94 - 50 = 44$ percentage points
Chatbot B	72%	Yes	$72 - 50 = 22$ percentage points
System C	58%	No	$58 - 50 = 8$ percentage points

Therefore, ELIZA and Chatbot B satisfy the 70% criterion as stated in the problem; System C does not.

(c) Improvement in fooling rate

ELIZA fooling rate = 6%; System C = 42%.

Improvement in percentage points = $42 - 6 = 36$ percentage points.

Ratio = $42 / 6 = 7$. Therefore, System C's fooling rate is 7 times ELIZA's rate (a 600% increase relative to ELIZA's 6% baseline).

Chatbot B → System C: $42 - 28 = 14$ percentage points improvement.

(d) Interpretation

The **Turing Test** measures whether a machine can act like a human during a conversation. A high fooling rate means that judges often mistake the machine for a human, but it **does not prove that the machine is actually thinking or acting rationally**.

According to the **Chinese Room argument**, a system may produce human-like answers by following rules for manipulating symbols without actually understanding their meaning. Therefore, even if **System C has a high fooling rate**, we cannot conclude that it truly understands, thinks, or reasons like a human.

Question 4: Rational Route Choice under Uncertainty

Topic: Concept of rationality (expected performance, rational versus omniscient agent) | 15 minutes

A rational taxi agent must reach the airport. The deadline is 55 minutes. Its performance measure (utility) for a trip is:

$$U = 100 - T - 20 \cdot L$$

where T is the travel time in minutes and $L = 1$ if the taxi arrives after the 55-minute deadline, otherwise $L = 0$.

Route	Normal time (min)	Probability of disruption	Time if disrupted (min)
R1: Highway	30	0.30 (accident)	70
R2: City road	45	0.10 (signal failure)	60
R3: Ring road	50	0 (never disrupted)	50

- (a) Compute the utility of each route in each outcome (normal and disrupted), clearly marking the late trips. **[4 Marks]**
- (b) Compute the expected utility of R1, R2 and R3 and state which route the rational agent should choose. **[5 Marks]**
- (c) A new percept arrives: a radio alert reports a major accident on the highway, raising the probability of disruption on R1 to 0.80. Recompute the expected utility of R1 and re-decide the route. **[4 Marks]**
- (d) Suppose the agent chose the route in part (b) and the disruption did occur. Compute the utility actually obtained and state whether the agent was irrational. Distinguish between a rational agent and an omniscient agent. **[4 Marks]**
- (e) Interpret: explain why rationality depends on the percept sequence and the performance measure, and why part (c) changes the decision although the roads themselves did not change. **[3 Marks]**

(a) Utility in each outcome

Route	Outcome	T (min)	L	Utility $U = 100 - T - 20L$
R1 – Highway	Normal	30	0	$100 - 30 = 70$
R1 – Highway	Disrupted / late	70	1	$100 - 70 - 20 = 10$ (LATE)
R2 – City road	Normal	45	0	$100 - 45 = 55$
R2 – City road	Disrupted / late	60	1	$100 - 60 - 20 = 20$ (LATE)
R3 – Ring road	Normal / certain	50	0	$100 - 50 = 50$

R3 has no disruption, so its utility is always 50.

(b) Expected utility and rational choice

Formula: $EU = P(\text{normal}) \times U(\text{normal}) + P(\text{disruption}) \times U(\text{disrupted})$.

R1: $EU = 0.70(70) + 0.30(10) = 49 + 3 = 52$.

R2: $EU = 0.90(55) + 0.10(20) = 49.5 + 2 = 51.5$.

R3: $EU = 1.00(50) = 50$.

Route	Expected utility
R1	52
R2	51.5
R3	50

Rational choice: R1 (Highway), because it has the highest expected utility of 52.

(c) New percept: highway disruption probability = 0.80

Updated R1 expected utility:

$EU(R1) = 0.20(70) + 0.80(10) = 14 + 8 = 22$.

Route	Expected utility after alert
R1 – Highway	22
R2 – City road	51.5
R3 – Ring road	50

New rational choice: R2 (City road), because 51.5 is now the highest expected utility. The new percept changes the probability of disruption, so the rational decision changes even though the physical routes have not changed.

(d) Actual outcome and rational versus omniscient agent

The route selected in part (b) was R1. If the disruption occurs, $T = 70$ and $L = 1$.

Actual utility $= 100 - 70 - 20(1) = 10$.

(e) Interpretation

Rationality depends on the percept sequence available to the agent and the performance measure used to evaluate its actions. The taxi agent chooses the route that gives the highest expected utility based on the information it currently has.

the radio alert provides new information that the probability of disruption on R1 has increased to 0.80. Therefore, the expected utility of R1 decreases, even though the actual road conditions and routes have not changed.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary (100% of marks)	Level 3 – Proficient (75%)	Level 2 – Developing (50%)	Level 1 – Beginning (25%)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Q. No. / Criteria	Problem Understanding (4)	Method Selection (4)	Solution Process (5)	Accuracy of Calculation (4)	Interpretation of Results (3)	Sub. Total (20)	Total %
1							
2							
3							
4							
Total							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____